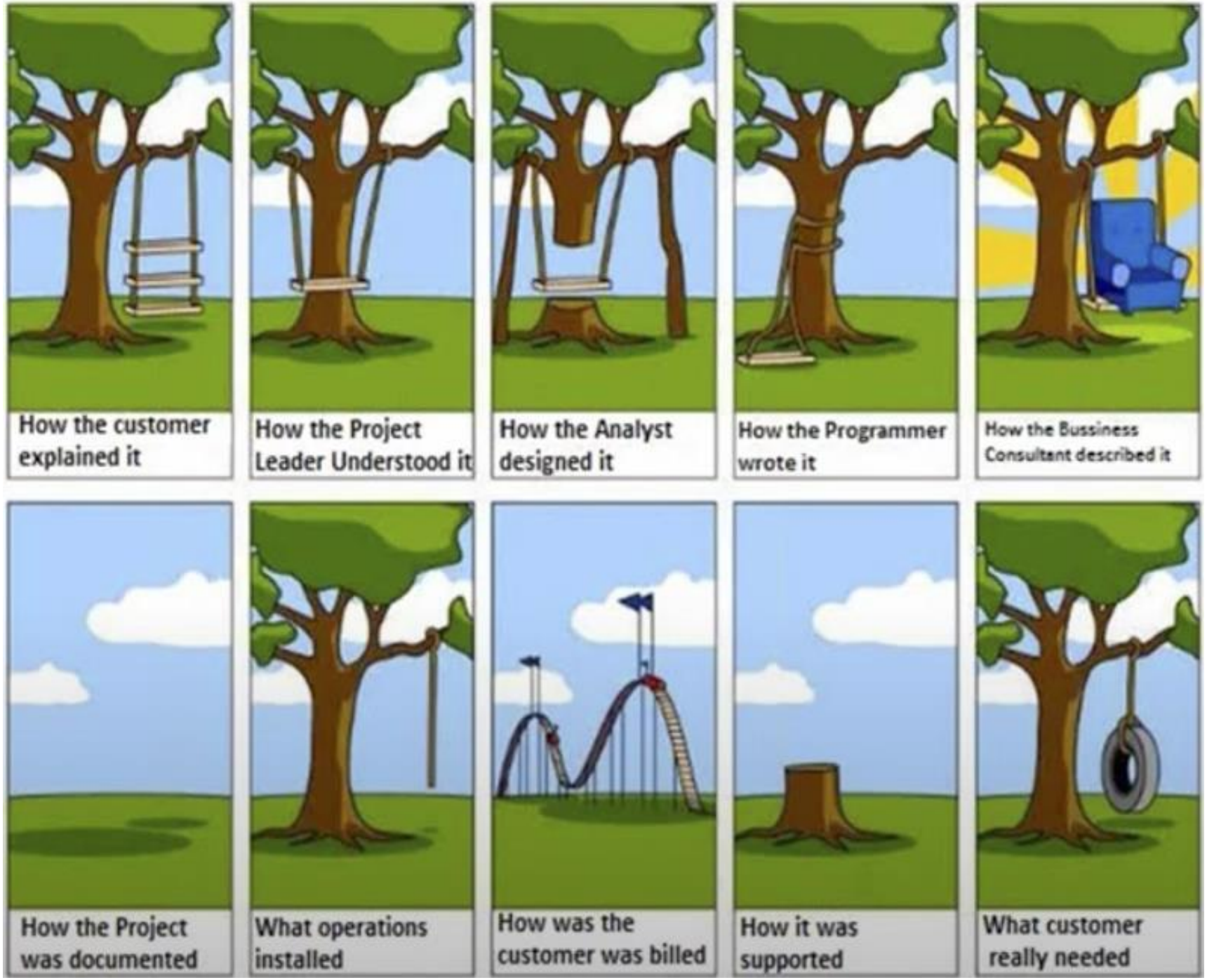
Project scope management

Scope

- Project scope management is one of the most critical project management knowledge (*skill*) areas
- Scope defines
  - *All* the work that is required to complete the project successfully and
  - *Only* the work that is required, no more, no less
- This latter point is critical: project scope management defines and controls what ***is*** and ***is not*** part of the project work


How the customer explained it
How the Project Leader Understood it
How the Analyst designed it
How the Programmer wrote it
How the Bussiness Consultant described it
How the Project was documented
What operations installed
How was the customer was billed
How it was supported
What customer really needed

Scope planning

- Scope planning is concerned with choosing the most appropriate, most effective approach to managing the scope of the project
- Scope planning defines how to:
  1. Define the scope
  2. Develop the detailed project scope statement
  3. Define and develop the WBS
  4. Verify project scope
  5. Control project scope
- Scope planning takes two primary inputs: the project charter and the preliminary project scope statement
- Scope planning results in a *project scope management plan*

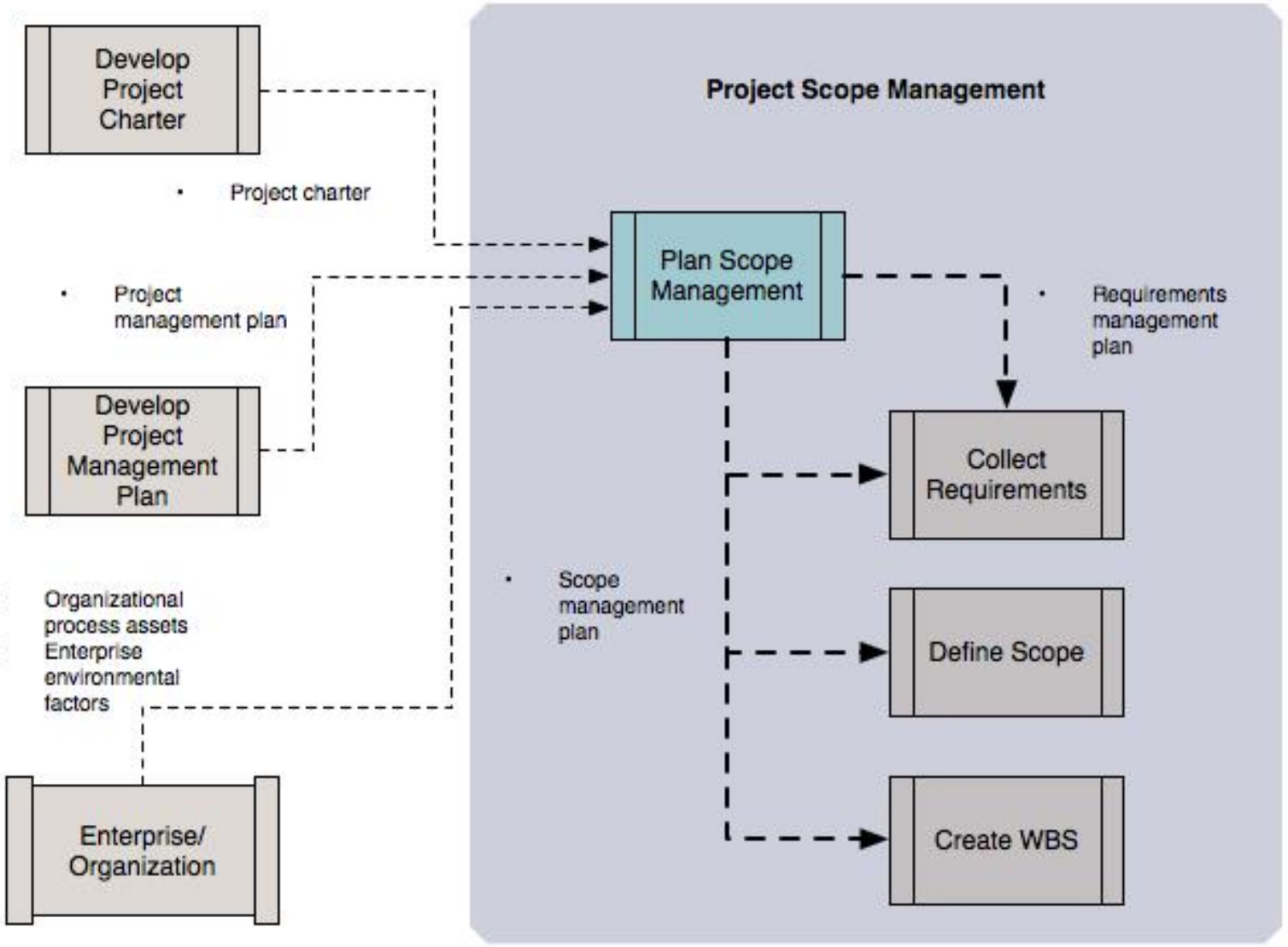
Scope management plan

- According to PMBOK 5: "The scope management plan can be formal **or** informal, broadly framed **or** highly detailed, based on the needs of the project."
  - This course adopts the informal, broadly-framed perspective
- The components of a scope management plan include:
  1. Process for preparing a detailed project scope statement
  2. Process that enables the creation of the WBS from the detailed project scope statement
  3. Process that establishes how the WBS will be maintained and approved
  4. Process that specifies how formal acceptance of the completed project deliverables will be obtained
  5. Process to control how requests for changes to the detailed project scope statement will be processed—this process is directly linked to the Perform Integrated Change Control process

Requirements management plan

- Components of the requirements management plan can include, but are not limited to:
  - How requirements activities will be planned, tracked, and reported
  - Configuration management activities such as:
    - How changes to the product will be initiated
    - How impacts will be analyzed, how they will be traced, tracked, and reported
    - Authorization levels required to approve these changes
  - Requirements prioritization process
  - Product metrics that will be used and the rationale for using them
  - Traceability structure to reflect which requirement attributes will be captured on the traceability matrix

Plan Scope Management: Data Flow Diagram



Process: Collect Requirements

- Recall the most critical lesson from the Standish Group's 2009 CHAOS report:
  - **Requirements, requirements, requirements**
    - » Requirements are unclear, incomplete, or the project management methodology does not accommodate changing requirements effectively
- Collecting initial requirements is a critical first step in managing requirements over the course of the project
  - In an iterative/incremental or adaptive/agile project, requirements collection continues throughout the life cycle of the project

Types of requirements

- ***Business requirements***. Describe the high-level needs of the organization as a whole
  - Examples: *Increase e-commerce purchases by 25%*
- ***Stakeholder requirements***. Describe the needs of a particular stakeholder, especially with respect to external stakeholders
  - Example: *As the site owner, I want the site supply chain applications to interface with the Web site*
- ***Functional requirements***.* Describe the behavior of the product
  - Example: *As a retail customer, I want to be able to shop by brand or product category*
- ***Nonfunctional requirements***.* Describe the behavioral qualities required for the product to be effective
  - Example: *As the Web site owner, I want the site to be available 99.99% of the time over a 1-year period*

*PMBOK 5 groups functional and nonfunctional requirements under the heading 'Solution Requirements'

Types of requirements

- ***Transition requirements***. Describe temporary capabilities, such as data conversion and training requirements, needed to transition from the current to the future state
  - *Example*: *As a data entry clerk, I want to be able to use the keyboard shortcuts from the previous order system, so that I can maintain my level of productivity*
- ***Project requirements***. Describe the actions, processes, or other conditions the project needs to meet
  - *Example*: *Project must use the hybrid plan-driven, iterative and agile methodology*
- ***Quality requirements***. Capture condition or criteria needed to validate the successful completion of a project deliverable
  - *Example*: *As the site owner, I want a retail customer test group to be able to successfully search for and find a requested product within 30 seconds*

Process: Create WBS

- **WBS –** Work Breakdown Structure. Technique for describing all work in a project.
- **PERT** – Program Evaluation and Review Technique. A well-entrenched technique for scheduling.
- **CPM** – Critical Path Method. Used with PERT to determine problems in scheduling.
- **Gantt Charts** – bar chart that graphically displays project schedule

The Work Breakdown Structure

- The **Work Breakdown Structure (WBS)** is a hierarchical description of the work that must be done to complete the project as defined in the Project Overview Statement (POS).
- The WBS terms
  - ***Activity***: An activity is simply a chunk of work.
  - ***Task*****:** A task is a smaller chunk of work.
  - ***Milestone***: a task of zero duration. Usually an external event. Can be used to mark completion of an activity or phase.

Decomposition

- Decomposition is usually performed in a top-down, hierarchical manner as expressed by the Work Breakdown Structure (WBS)
- Creating the *Work Breakdown Structure* (WBS) is the process of decomposing project deliverables and work into smaller, more manageable components

Work Breakdown Structure

- A Work Breakdown Structure (WBS) captures all the work of a project in an organized way.
  - Portrayed graphically as a hierarchical tree,
  - A tabular list of "element" categories and tasks or the indented task list that appears in your Gantt chart schedule.
- Breaking Large, complex projects into progressively smaller pieces
  - A collection of defined "work packages" that may include a number of tasks.
  - Continue to refine work until packages are of a suitable size
- A work package can include design, coding, testing, etc.

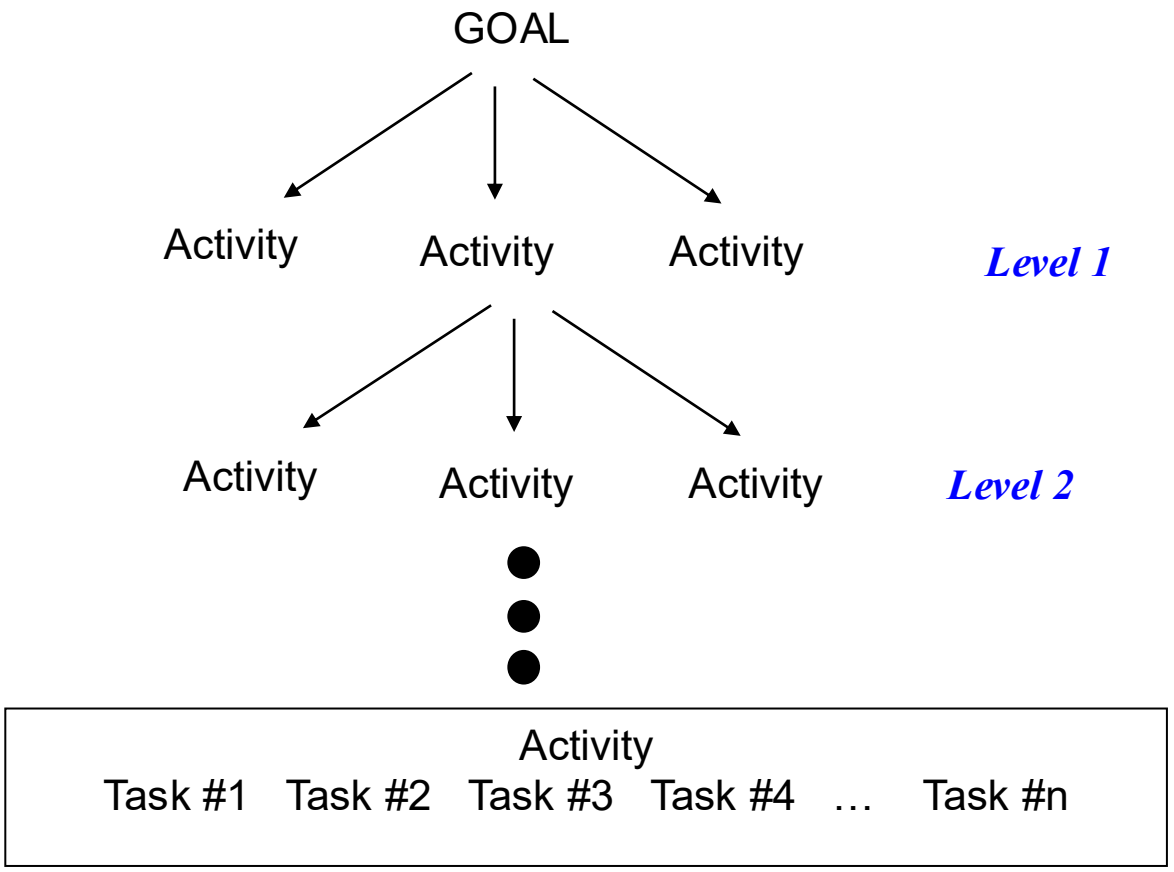
Decomposition activities

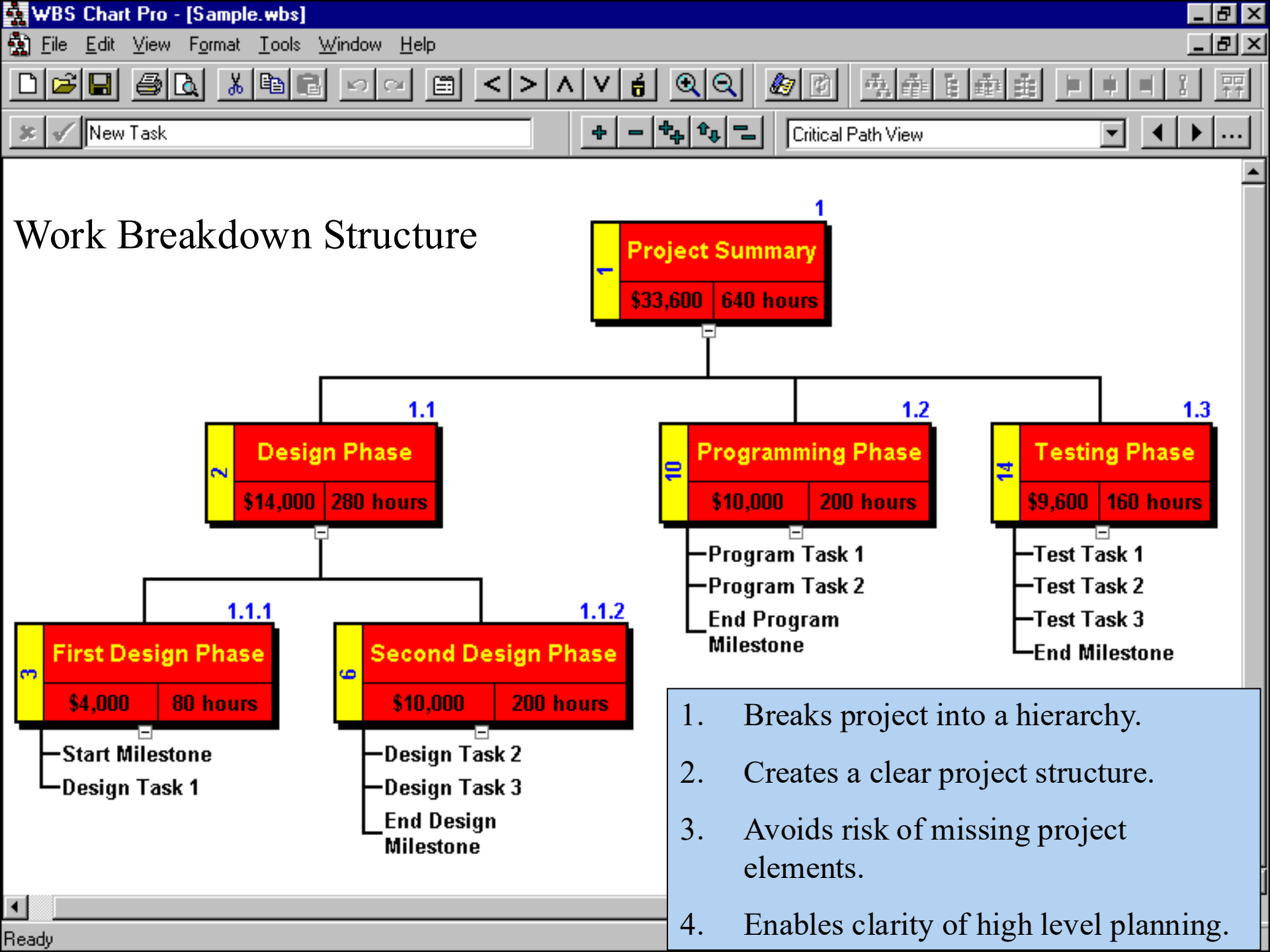
- Identify and analyze the deliverables and related work
  - The project scope statement provides the basis for the first iteration of deliverable identification
- Structure and organize the WBS according to an appropriate high-level hierarchy
  - For IT projects using an iterative system development model, a phase/iteration/discipline hierarchy most closely matches the SDLC process
- Decompose the upper WBS levels into lower-level detailed components
  - If appropriate, larger deliverables may be decomposed into smaller deliverables, all the way down to the work package level

Decomposition activities

- Develop and assign identification codes to the WBS components
  - Note that each item in the WBS has both a unique identifier (the *code or account* identifier) and a WBS code
  - The unique identifier does not change if a component is moved about in the WBS structure; however, the **WBS code** does change
- Verify that the amount of decomposition of work is necessary and sufficient
  - This can be translated into a simple concept: the 'just right' principle
  - The just right principle states that no more decomposition is needed–it would be too detailed–and any less decomposition would be too coarse
  - Insufficient decomposition leads to reduced ability to plan, manage, and control the work
  - Excessive decomposition leads to wasted effort and decreased efficiency

The Work Breakdown Structure




WBS Chart Pro - [Sample.wbs]
File Edit View Format Tools Window Help
New Task
Critical Path View
Work Breakdown Structure
1
1 Project Summary
$33,600 640 hours
1.1
2 Design Phase
$14,000 280 hours
1.2
10 Programming Phase
$10,000 200 hours
1.3
14 Testing Phase
$9,600 160 hours
Program Task 1
Program Task 2
End Program Milestone
Test Task 1
Test Task 2
Test Task 3
End Milestone
1.1.1
3 First Design Phase
$4,000 80 hours
Start Milestone
Design Task 1
1.1.2
6 Second Design Phase
$10,000 200 hours
Design Task 2
Design Task 3
End Design Milestone
1. Breaks project into a hierarchy.
2. Creates a clear project structure.
3. Avoids risk of missing project elements.
4. Enables clarity of high level planning.
Ready

Six Criteria to Test for Completeness in the WBS

- The WBS is developed as part of a Joint Planning session. But how do you know that you've done this right? Each activity must possess six characteristics to be considered complete – that is, completely decomposed. The six characteristics are
    1. Status/completion is measurable
    2. Start/end events are clearly defined
    3. Activity has a deliverable
    4. Time/cost is easily estimated
    5. Activity duration is within acceptable limits
    6. Work assignments are independent

    » Let us review each one in detail …

Six Criteria to Test for Completeness in the WBS

- **Measurable Status:** The project manager can ask for the status of an activity at any point in time during the project. If the activity has been defined properly, that question is answered easily.
  - Example: a system's documentation is estimated to be about 300 pages long and requires approximately four months of full-time work to write, here is a possible report that a developer can provide regarding the status:

    "I've written 150 pages, so I guess I am 50 percent complete."

Six Criteria to Test for Completeness in the WBS

- **Bounded with Start/end events** :
  - Each activity should have a clearly defined start & end event.
  - Once the start event has occurred, work can begin on the activity.
  - The deliverable is most likely the end event that signals work is closed on the activity.
  - **Example**: using the systems documentation example: The **start event** might be notification to the team member who will manage the creation of the systems documentation that the final acceptance tests of the systems are complete. The **end event** would be notification to the project manager that the customer has approved the systems documentation.

Six Criteria to Test for Completeness in the WBS

- **Deliverable**
  - The result of completing the work that makes up the activity is the production of a deliverable.
  - The deliverable is a visible sign that the activity is complete.
  - This could be an approving manager's signature, a physical product or document.
- **Cost/Time Estimate**
  - Each activity should have an estimated time and cost of completion.
  - Being able to do this at the lowest level of decomposition in the WBS allows you to aggregate to higher levels and the total project cost and the completion date.

Six Criteria to Test for Completeness in the WBS

- **Activity Independence**
  - It is important that each activity be independent. Once work has begun on the activity, it can continue reasonably without interruption and without the need of additional input or information until the activity is complete.
  - Though it is possible that an activity could be scheduled during different times based on resource availability.

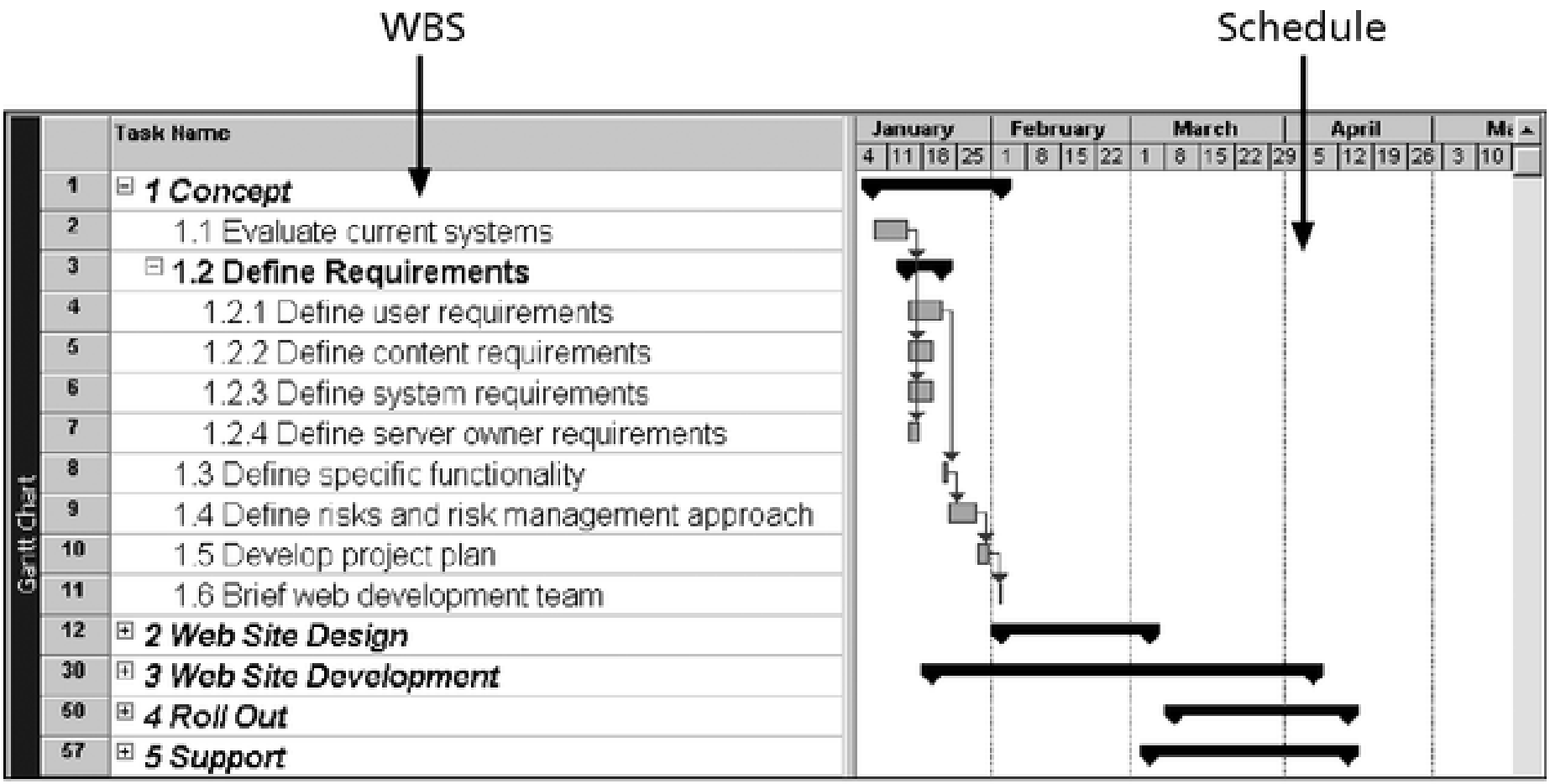
Approaches to Building the WBS

- There are many ways to build the WBS. There is no one correct way to create the WBS.
- Hypothetically, if we put each team member in a different room and asked that person to develop the project WBS, they might all come with different answers.
- There are three general approaches to building the WBS:
  1. **Noun-type** approaches.
  2. **Verb-type** approaches.
  3. **Organizational** approaches

Approaches to Building the WBS

- **Noun-type approaches**. Noun-type approaches define the deliverable of the project work in terms of the components (*physical* or *functional*) that make up the deliverable.
- **Verb-type approaches**. Verb-type approaches define the deliverable of the project work in terms of the actions that must be done to produce the deliverable. These include the *design-build-test-implement* and project *objectives* approaches.
- **Organizational approaches.** Organizational approaches define the deliverable of the project work in terms of the organizational units that will work on the project. This type of approach includes the department, process, and geographic location approaches.

WBS and Gantt Chart in Microsoft Project



The 100% rule

- The 100% rule is one of the most important principles guiding development, decomposition, and evaluation of the WBS
- Rule states that the WBS:
  - Includes 100% of the work defined by the project scope
  - Captures all internal, external, and interim deliverables in terms of work to be completed, including project management
- Rule applies at all levels within the hierarchy: the sum of the work at the "child" level must equal 100% of the work represented by the "parent"
- WBS should not include any work that falls outside the actual scope of the project, that is, it cannot include more than 100% of the work

Conventional WBS levels

- Level 1
  - Project name
- Level 2
  - Major subsystems of the project
- Level 3
  - Components/task activities of subsystems at Level 2
- Level 4
  - Subcomponents/subtasks of components/tasks at Level 3
- Level 5
  - Work packages for subcomponents/subtasks at Level 4
  - Work packages are where the actual work takes place
  - Assigned to a person and given a schedule and budget

Note that conventional WBS decomposition levels are *product-oriented*

Sample conventional WBS

Note early commitment to a specific system decomposition or architecture

Comparison of different projects requires extracting this information for each subsystem & combining

Other subsystems have same WBS pattern as sensor detection subsystem

	WBS	Task Name
1	**1**	**− Automatic Toll System**
2	**1.1**	**− System design**
3	1.1.1	Architectural design
[illegible]	**1.2**	**− Toll booth subsystem**
5	**1.2.1**	**− Sensor detection subsystem**
6	**1.2.1.1**	**− Requirements**
7	1.2.1.1.1	Requirements document
8	**1.2.1.2**	**− Analysis**
[illegible]	[illegible]	Domain model
10	1.2.1.2.2	Use-case model
11	**1.2.1.3**	**− Design**
12	1.2.1.3.1	Component design
13	**1.2.1.4**	**− Implementation**
14	1.2.1.4.1	Test case development
15	1.2.1.4.2	Coding
16	**1.2.1.5**	**− Integration**
[illegible]	1.2.1.5.1	Integration test cases
18	[illegible]	**+ Toll Authority interface**
24	**1.3**	**+ Toll Authority subsystem**

Create WBS: Outputs

- *Scope baseline*. The scope baseline is the approved version of a scope statement, work breakdown structure (WBS), and its associated WBS dictionary:
  - *Project scope statement*. Output from Define Scope process
  - *WBS*. Either graphical or tabular form
  - *WBS dictionary*. A companion document to the WBS, providing detailed information about components in the WBS, including work packages

WBS dictionary content

- **Required**
  - *Code or account identifier* – unique number assigned to a WBS component
  - *Statement of work* – scope statement for the WBS component
  - *Responsible organization* for WBS component
  - *Schedule milestones* for WBS component
- **Optional**
  - *Contract information*
  - *Quality requirements* – may be used in assessment planning
  - *Technical references* to assist in executing work
  - *Charge number*
  - *List of schedule activities*
  - *Resources required*
  - *Estimate of cost*

WBS template

Component groups with a '+' in front of them are 'rolled up' – subcomponents are hidden to reduce clutter

	WBS	Name
1	**1**	− **Project**
2	**1.1**	+ **Management**
20	**1.2**	+ **Environment**
33	**1.3**	+ **Requirements**
42	**1.4**	− **Design**
43	1.4.1	Inception phase architecture prototyping
44	**1.4.2**	− **Elaboration phase architecture baselining**
45	1.4.2.1	Architecture design modeling
46	1.4.2.2	Design demonstration planning and conduct
47	1.4.2.3	Software architecture description
48	**1.4.3**	+ **Construction phase design modeling**
51	1.4.4	Transition phase architecture design maintenance
52	**1.5**	− **Implementation**
53	1.5.1	Inception phase component prototyping
54	**1.5.2**	− **Elaboration phase component implementation**
55	1.5.2.1	Critical component coding demonstration integration (architectural baseline)
56	**1.5.3**	+ **Construction phase implementation**
61	1.5.4	Transition phase component maintenance
62	**1.6**	− **Assessment**
63	1.6.1	Inception phase assessment planning
64	**1.6.2**	− **Elaboration phase assessment**
65	1.6.2.1	Test modeling
66	1.6.2.2	Architecture test scenario implementation
67	1.6.2.3	Demonstration assessment and release descriptions
68	**1.6.3**	+ **Construction phase assessment**
72	**1.6.4**	+ **Transition phase assessment**
74	**1.7**	+ **Deployment**

Activity or task definition

	WBS	Name
1	**1**	**– Automatic Toll Booth System**
42	**1.4**	**– Design**
43	1.4.1	Inception phase architecture prototyping
44	**1.4.2**	**– Elaboration phase architecture baselining**
45	**1.4.2.1**	**– System architecture definition**
[illegible]	[illegible]	Define high-level hardware architecture
47	1.4.2.1.2	Define high-level communications architecture
48	**1.4.2.2**	**– Architecture design modeling**
49	1.4.2.2.1	Perform ABD (architecture-based design)
50	[illegible]	Perform quality attribute analysis (ATAM)
[illegible]	1.4.2.2.3	Perform cost/benefit analysis (CBAM)
[illegible]	1.4.2.2.4	Document software architecture
53	**1.4.2.3**	**– Design demonstration planning and conduct**
54	1.4.2.3.1	Select architectural baseline critical use-cases
55	1.4.2.3.2	Select architectural baseline critical functionality
56	[illegible].2.3.3	Design architectural baseline components
[illegible]	**1.4.3**	**+ Construction phase design modeling**
60	1.4.4	Transition phase architecture design maintenance
61	**1.5**	**– Implementation**
62	1.5.1	Inception phase component prototyping
63	**1.5.2**	**– Elaboration phase component implementation**
64	**1.5.2.1**	**– Critical component coding demonstration integration (architectural baseline)**
65	1.5.2.1.1	Implement architectural baseline components
66	1.5.2.1.2	Integrate architectural baseline components
[illegible]	**1.5.3**	**+ Construction phase implementation**
[illegible]	1.5.4	Transition phase component maintenance
73	**1.6**	**– Assessment**
74	1.6.1	Inception phase assessment planning
75	**1.6.2**	**– Elaboration phase assessment**
76	1.6.2.1	Select and define architectural baseline tests
77	1.6.2.2	Implement architecture test scenario
78	1.6.2.3	Test architectural baseline

Added *System architecture definition* WBS component

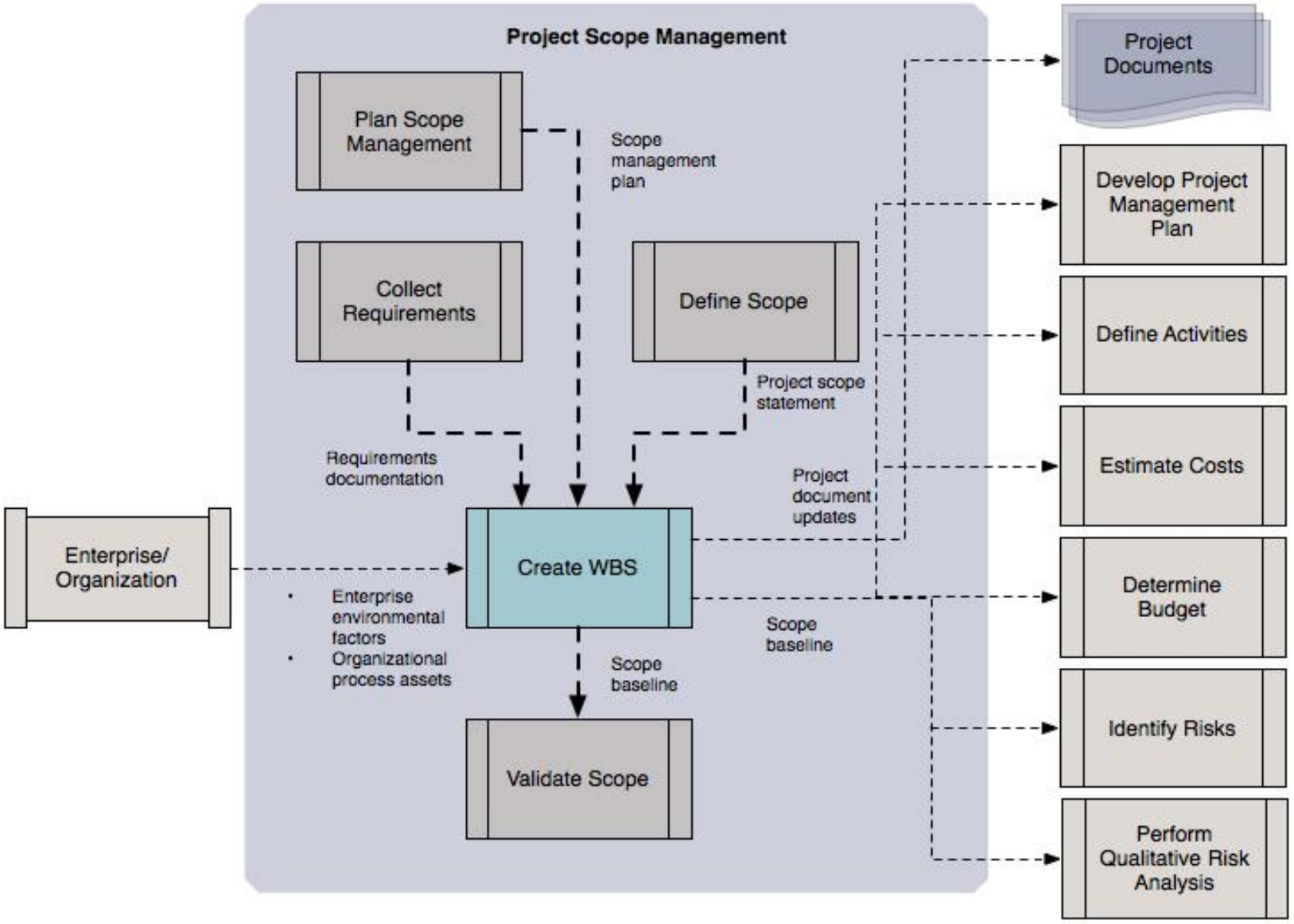
Note expansion and detailing of WBS template *Architecture design modeling* entry; renamed *Software architecture description* to *Document software architecture*

Note expansion and detailing of WBS template *Design demonstration planning and conduct* entry

Note expansion and detailing of WBS template *Critical component coding demo integration* entry

Note rework of WBS template *Elaboration phase assessment* entry

Create WBS: Data Flow Diagram



Agile Perspective: Create WBS

Valuable concepts and tools

- The *Create WBS* process is among the least compatible with a complex (or agile) project perspective
  - ➢ It encourages solidifying the scope of the project in a complex, difficult to change artifact, the WBS
  - ➢ Once an artifact is created, it assumes an authority that may not be justified
  - ➢ Most WBSs are out-of-date shortly after being created
  - ☛ People are reluctant to toss something that has taken so much effort
- Nevertheless, the process of thoughtful decomposition of the product into smaller, more manageable pieces is certainly compatible with an adaptive/agile methodology
- Adaptive/agile limits high-level decomposition to the product roadmap, and defers low-level decomposition to the release and iteration (sprint) level

WBS – Basis of Many Things

- Network scheduling
- Costing
- Risk analysis
- Organizational structure
- Control
- Measurement

The MS-Project Process

- Move WBS into a Project outline (in Task Sheet)
- Add resources (team members or roles)
- Add costs for resources
- Assign resources to tasks
- Establish dependencies
- Refine and optimize
- Create baseline
- Track progress (enter actuals, etc.)

Defining Task Sets

- Determine type of project
- Assess the degree of rigor required
- Identify adaptation criteria
- Select appropriate software engineering tasks
- Task Set Refinement
  - Use Work Breakdown Structure to determine tasks
- Define a Task Network
  - Use a Gantt Chart and/or PERT to document